

Federated Generative AI Framework

for Glaucoma Clinical Translation

Concise Research Framework

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Research Objective

Develop a locally-deployable generative AI system for glaucoma clinical decision support that combines uncertainty-aware image analysis (H-UQ-MFF) with large language models to provide clinically-validated recommendations, explanations, and confidence assessments using EyePACS-AIROGS-light-V2 foundation.

Core Problem

Current glaucoma AI systems provide binary classifications without clinical context, explanations, or uncertainty quantification. Clinical decision support lacks:

- Natural language integration of multimodal data (images + clinical notes + patient history)
- Uncertainty-aware recommendations for safe clinical deployment
- Contextual reasoning that matches ophthalmologist decision-making patterns
- Local deployment capability for HIPAA-compliant healthcare environments

No existing framework combines foundation vision models with clinical LLMs for uncertainty-aware glaucoma management.

Implementation Plan

Phase 1: Multimodal Architecture Development

- Extend H-UQ-MFF framework to generate structured clinical features with uncertainty
- Integrate local medical LLM (MedGemma-4B/BioMistral-7B) with vision pipeline
- Develop uncertainty propagation from vision model to language generation
- Create clinical reasoning templates for glaucoma assessment and management

Phase 2: Clinical Integration & Validation

- Generate clinical scenarios and synthetic patient histories for EyePACS-AIROGS-light-V2
- Cross-dataset validation on REFUGE/PAPILA with explanatory outputs
- Expert ophthalmologist evaluation of AI-generated clinical recommendations
- Uncertainty calibration across demographic groups and imaging conditions

Phase 3: Deployment & Regulatory Framework

- Local deployment optimization using GGUF quantization and llama.cpp
- FDA-compliant evaluation protocol for generative AI medical devices
- Clinical utility assessment in simulated healthcare workflows
- Documentation framework for LLM-based clinical decision support approval

Essential Datasets & Resources

Primary Dataset Access Links

1. **EyePACS-AIROGS-light-V2:** <https://github.com/thinhdao276/EyePACS-AIROGS-light-V2>
2. **REFUGE Challenge:** <https://refuge.grand-challenge.org/>
3. **PAPILA Dataset:** <https://figshare.com/articles/dataset/PAPILA/14798004>
4. **RETFound Model:** https://github.com/rmaphoh/RETFound_MAE
5. **MedGemma-4B:** <https://huggingface.co/google/medgemma-4b-it>
6. **BioMistral-7B:** <https://huggingface.co/BioMistral/BioMistral-7B>
7. **GRAPE Dataset:** <https://figshare.com/articles/dataset/GRAPE/21392197>
8. **LLaVA-Med:** <https://github.com/microsoft/LLaVA-Med>
9. **MedSpaCy:** <https://github.com/medspacy/medspacy>
10. **llama.cpp:** <https://github.com/ggml-org/llama.cpp>

Technical Implementation Guidelines

Architecture Specifications

- **Vision Encoder:** H-UQ-MFF with Monte Carlo Dropout for epistemic uncertainty
- **Vision-Language Bridge:** 2-layer MLP projection (GELU activation)
- **Language Model:** MedGemma-4B or BioMistral-7B with QLoRA fine-tuning
- **Local Deployment:** GGUF quantization (Q4_K_M) for consumer GPU deployment (8-24GB VRAM)

Expected Deliverables

Technical Outputs

11. **Open-Source Framework:** Complete pipeline for uncertainty-aware glaucoma clinical decision support
12. **Model Artifacts:** Fine-tuned vision-language model with quantized deployment package
13. **Deployment Guide:** Documentation for local healthcare system implementation

Academic Publications

14. **Primary Paper:** "Federated Generative AI for Glaucoma Clinical Decision Support" (Target: Nature Medicine/NEJM AI)
15. **Technical Paper:** "Local Deployment of Medical Vision-Language Models with Safety Guarantees" (Target: IEEE TBME)
16. **Clinical Validation:** "Clinical Utility of AI-Generated Explanations in Glaucoma Management" (Target: Ophthalmology)

Unique Value Proposition

- **First uncertainty-aware generative AI** for glaucoma clinical decision support
- **Local deployment capability** ensuring patient privacy and regulatory compliance
- **H-UQ-MFF advantage** providing proven uncertainty quantification framework
- **FDA-compliant development** from project inception with regulatory pathway innovation

This framework establishes the foundation for revolutionary clinical decision support that combines technical innovation with rigorous clinical validation and regulatory compliance, positioning federated generative AI as the next frontier in ophthalmology practice.